Two reducible parameters disprove Conjecture 7753

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Abstract

The claimed bound of 2^{n-1} on the number of parameters for which the dynatomic polynomial is reducible fails already at $n = 1$. The two distinct integer parameters $c = -2$ and $c = -6$ give nontrivial factorizations over $\mathbb{Q}$. The accompanying self-contained Lean 4 project proves the polynomial identities, nonunit status of the factors, and failure of the cardinality bound.

1 The clause being disproved

Conjecture 00000007753 defines $f_c(X) = X^2 + c$ and

$$\Phi_{n,c}(X) = \prod_{d|n} (f_c^{\circ d}(X) - X)^{\mu(n/d)}.$$

Its final clause asserts that, for fixed n, the exceptional parameters c making this polynomial reducible are finite in number and number at most 2^{n-1}. Here reducibility is over $\mathbb{Q}$; the witnesses below are integers and therefore also rational parameters. No restriction on c excluding either witness, and no restriction $n > 1$, occurs in the statement.

We disprove that bound for the original polynomial in X. We do not use, or need to interpret, the separate claim about the Galois group of the special value at $X = 0$.

2 Counterexample

Since the only positive divisor of 1 is 1 and $\mu(1) = 1$,

$$\Phi_{1,c}(X) = f_c(X) - X = X^2 - X + c.$$

Consequently,

$$\Phi_{1,-2}(X) = X^2 - X - 2 = (X - 2)(X + 1),$$
$$\Phi_{1,-6}(X) = X^2 - X - 6 = (X - 3)(X + 2).$$

Each polynomial on the left is nonzero, and each factor on the right is a nonunit in $\mathbb{Q}[X]$. Indeed, all four factors have degree one. Alternatively, each has a rational root, so if it had a polynomial inverse, evaluation at that root would give $0 = 1$.

Theorem 1. *Conjecture 00000007753 is false.*

Proof. Both -2 and -6 belong to the exceptional-parameter set at $n = 1$, and they are distinct. That set therefore has at least two elements. The asserted upper bound is $2^{1-1} = 1$, a contradiction. □

This argument requires no computation of Galois groups, dynatomic curves, or an exhaustive parameter search.

3 Formalization and semantic correspondence

The project in `lean/` uses Lean 4.31.0 and the standard library only; Mathlib and network dependencies are not required.

- `Poly` is a finite list of rational coefficients in ascending degree order. `mul` is ordinary polynomial convolution, implemented by scalar multiplication, coefficientwise addition, and a one-position shift. `trim` removes trailing zero coefficients. `EqPoly` is equality after this normalization, hence ordinary coefficient equality.

- `IsUnit p` means that there exists an arbitrary rational-coefficient polynomial q whose product with p normalizes to the constant polynomial 1. `Reducible p` asserts that p is nonzero and is the product of two nonunits. Thus reducibility is proved in the polynomial algebra, not merely by reporting a discriminant or evaluating a few points.

- `eval_mul` and `eval_trim` prove, for every rational input and every coefficient list, that evaluation respects multiplication and normalization. `root_not_unit` uses these general theorems to rule out every possible polynomial inverse of a polynomial with a rational root.

- `firstDynatomic` is $f_c - X$, the defining divisor-product formula at $n = 1$. Its coefficient theorem, `firstDynatomic_coeffs`, proves its exact coefficient list is $[c, -1, 1]$. `factorA`, `factorB`, `reducibleA`, and `reducibleB` verify the two complete factorizations and their hypotheses.

- `HasAtMost P k` means that a list of length at most k covers every element satisfying P. This is precisely a cardinality upper bound; allowing duplicate entries does not weaken that bound. The final theorem

  `TLMC7753.exceptional_parameter_bound_false`

  proves the negation of this bound for the exceptional-parameter set and $k = 2^{1-1}$.

The full conjecture includes claims beyond this clause. Formalizing a false necessary clause suffices to disprove the conjunction; the Lean theorem does not assert a result about the ambiguous special-value Galois-group clause.

4 Reproduction and trust boundary

With the specified toolchain on the path, run from `lean/`:

```
lake build
lake env lean Audit.lean
lake env leanchecker Main
```

The supplied `Audit.lean` prints the axioms used by the main theorems. They are exactly Lean's standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no custom axioms, proof holes, or native decision procedures. Closed rational arithmetic is checked using `decide +kernel`, which invokes kernel reduction, not native compilation.

The report can be rebuilt with `pdflatex -no-shell-escape solution.tex`. The optional `verify.py` independently checks the displayed coefficient products using Python's exact rational arithmetic; it is not part of the trusted proof.